

Community Detection Report

Citation network analysis

1. Results requested

This report summarizes community size, singleton communities, node and edge coverage, concentration in the largest communities, modularity, and measured runtime.

2. Community-size statistics

Statistic	Value
Minimum	1
Maximum	109,371,240
Median	3
Mean	438,279.888
25th percentile	1
75th percentile	17.5
90th percentile	212.6
99th percentile	29,150

3. Coverage and other metrics

Metric	Value
Total papers/nodes	109,569,972
Total citation edges	1,926,971,191
Number of communities	250
Singleton communities	91
Largest community fraction	99.8186%
Top 10 communities fraction	99.9865%
Node coverage	100%
Edge coverage	100%
Modularity Q	0.0000213681
Measured analysis runtime	70,379.24 seconds (19.55 hours)

4. Calculation methodology

Community sizes were calculated as $s_i = |\{v : C(v)=i\}|$, where $C(v)$ is the community assigned to paper v . The mean is the sum of all community sizes divided by the number of communities; the median and percentiles were calculated after sorting community sizes. Singleton communities were counted using sum of the indicator $1(s_i = 1)$. The largest-community fraction is $\max(s_i)/N$, and the top-10 fraction is the sum of the ten largest sizes divided by N . Node coverage is assigned nodes divided by N . Edge coverage is the number of citation edges whose two endpoints have valid community assignments divided by the total number of edges.

5. Modularity calculation

Modularity was calculated using $Q = \sum_c [l_c/M - (d_c/(2M))^2]$, where l_c is the number of citation edges internal to community c , d_c is the sum of in-degree and out-degree of nodes in c , and M is the total number of citation edges. The resulting modularity was $Q = 0.0000213681$. A value close to zero indicates weak separation relative to a degree-preserving random-network baseline.

6. Requested figures

The community-size distribution is plotted on logarithmic axes. The cumulative fraction of papers is plotted against communities ranked by decreasing size, with a logarithmic x-axis. The supplied figures show a highly

concentrated distribution: the cumulative curve reaches approximately 99.82% at the first community and remains nearly flat thereafter.

7. Runtime calculation

Runtime was measured as elapsed wall-clock time: `start_time = time.time(); analysis; runtime_seconds = time.time() - start_time`. The measured runtime was 70,379.24 seconds, approximately 19.55 hours. This is the analysis and modularity-calculation runtime, not confirmed original detection runtime.

8. Interpretation

The community partition is extremely imbalanced. The largest community contains nearly the entire network, while most remaining communities are very small. The modularity is close to zero, indicating weak separation relative to a degree-preserving null model. The high internal-edge fraction (99.1023%) is largely explained by the giant community and should not be interpreted alone as evidence of strong community structure.

9. Reproducibility note

Dataset: `public/data/citation_network.db`. The graph contains 109,569,972 nodes and 1,926,971,191 citation edges. Node and edge coverage are both 100%.